Helix Thready — Workable-Items Backlog (ATM-NNN)

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Rev	Date	Author	Change
1	2026-07-21	swarm (development)	Initial backlog — ATM-001… ATM-069 mapped to the four phases
2	2026-07-21	swarm (development, review)	Review pass — added ATM-070/071/072; completed the §9 gap-register coverage matrix (added 2.4, 2.8, 7.4, 8.3, 9.2, 9.4, 13) so the completeness claim holds
3	2026-07-22	swarm (development, pass 3)	Depth pass — added the full-granularity companion workable-items-detail.md; closed ATM-067 (AgentPool read at source); rescoped ATM-062 (session_orchestrator now shipped, not DESIGN-ONLY); partial-closed ATM-068 (`Agentic` verified); updated the §8 open-items register accordingly
4	2026-07-22	swarm (development, critic pass)	Linked the `test_types` field to the new canonical **15 mandated test types** enumeration in coding-standards.md §8 so each item's coverage is verifiable against a self-contained list

This document is the granular, decoupled, agent-implementable backlog for Helix Thready. Every item is an `ATM-NNN` ("**A**utonomous **T**hready **M**odule") workable item: independently scoped, with explicit dependencies, acceptance criteria, mandated test types, and `[GAP: …]` tags linking it to the private gap register. The backlog maps onto the four phases of final request §5.1.2. Items are the unit the orchestration model claims, dispatches and reviews (see agent-orchestration.md).

> **Source-of-truth note `[CONSTITUTION §11.4.93/95]`.** The authoritative store for workable items is a **SQLite DB tracked in git** at `docs/workable_items.db`, regenerated to/from this Markdown by `workable-items sync md-to-db`. This document is the human-readable projection; the DB is the machine source. The DB is **never gitignored** and is committed + pushed alongside the MD on every state change.

> **Full-granularity companion.** This file is the **summary** view (id/kind/pri/deps/gap tables). The **implementation-ready** per-item cards — task-level sub-items (`ATM-NNN.k`), Given/When/Then

> acceptance, explicit blocks/blocked-by edges, and `[VERIFIED-SOURCE]` anchors for every claim read at module source in Pass 3 — live in workable-items-detail.md.

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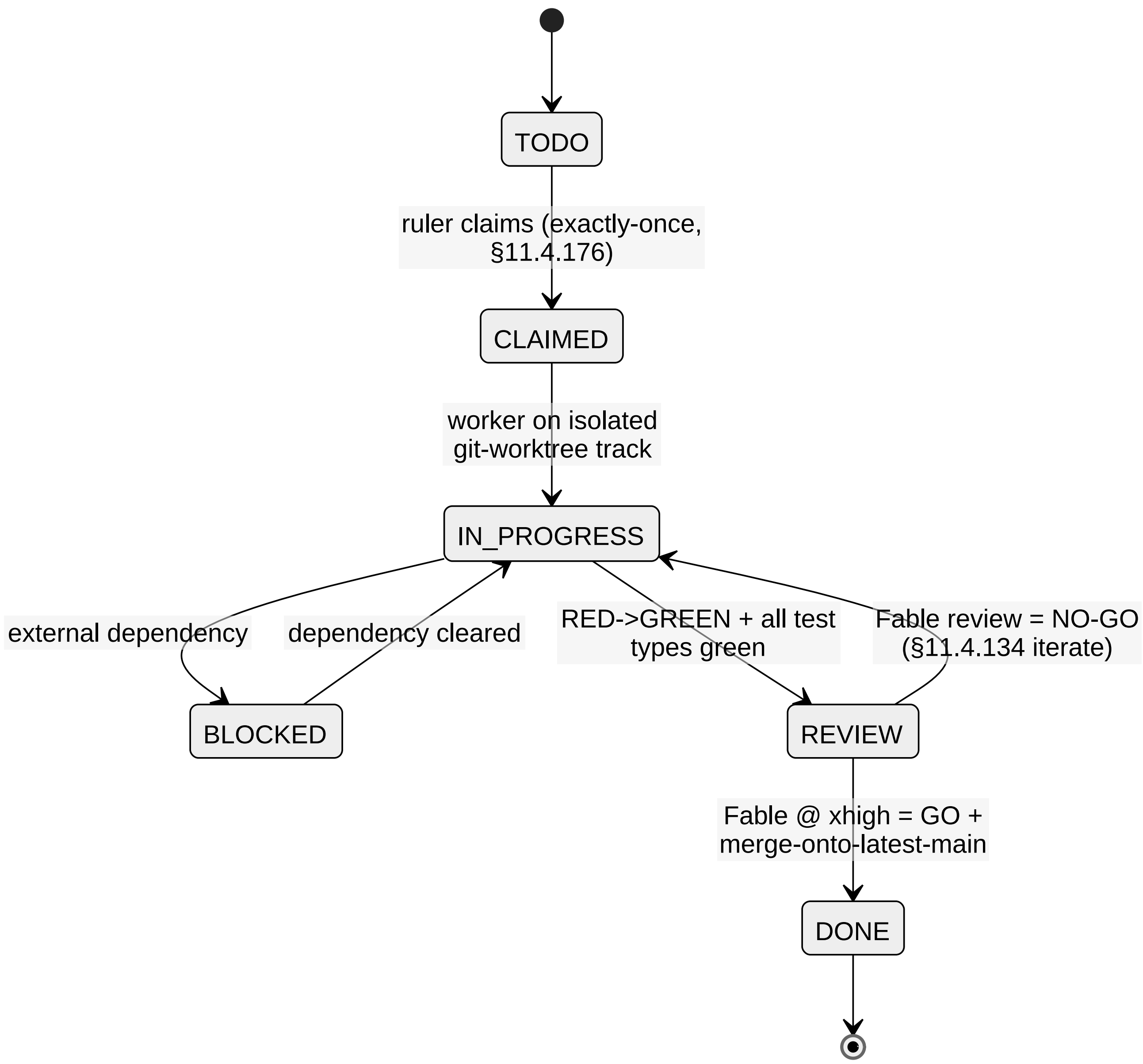
1. Item schema

Each item carries the fields below. The `id`, `phase`, `title`, `depends_on`, `test_types` and `gap` fields are mirrored 1:1 into the tracked SQLite DB.

```yaml
# One workable item as stored (projection of a row in docs/workable_items.db)
id: ATM-025
title: "Skill-dispatch engine (recipe-per-hashtag) on HelixSkills"
phase: "2.2"                    # phase.subphase from final request §5.1.2
kind: BUILD-NEW                 # PRODUCTION-WIRE | EXTEND | BUILD-NEW | HARDEN | AUDIT
owner_track: processing         # canonical multi-track domain (§11.4.191)
status: TODO                    # TODO | CLAIMED | IN_PROGRESS | REVIEW | DONE | BLOCKED
priority: P0                    # P0 blocks MVP | P1 GA | P2 hardening
depends_on: [ATM-011, ATM-023, ATM-036]
provides: "content-type -> Skill(s) resolution + ordered execution over BackgroundTasks"
decoupled_scope: |
  New submodule that maps hashtag/content-type -> Skill(s) from the helix_skills
  Skill-Graph, orders download->convert->analyze->research->reply, and runs each
  step via digital.vasic.background with the idempotent single-claim. Does NOT
  modify helix_skills' knowledge model (read-only consumer of the DAG).
acceptance_criteria:
  - "Given a post tagged #Video #Research, both the video-download Skill and the deep-research
      Skill run, download-first."
  - "A second delivery of the same post.received event does not double-dispatch (idempotent
      claim)."
  - "Unknown/insufficient tags fall through to indirect determination (ATM-024), never silently
      dropped."
test_types: [unit, integration, e2e, chaos, stress, performance, challenges, helixqa]
gap: ["4.1"]                    # gap-register subsystem ids addressed
provenance: ["IN-HOUSE: helix_skills", "BUILD-NEW", "OPERATOR: all-categories-parallel"]
review: "Fable @ xhigh (Opus xhigh fallback) — §11.4.209"
```

Field notes: `kind=PRODUCTION-WIRE` means “wire an existing production module by config”; `EXTEND` means “add capability to an existing owned module”; `BUILD-NEW` is a confirmed new-submodule gap; `HARDEN` raises a scaffold to reliance-grade; `AUDIT` is a re-verification/decoupling task. Every non-unit test type exercises the real system `[CONSTITUTION §11.4.27]`; mocks/stubs are unit-only. Each item’s `test_types` names the **applicable subset** of the canonical **15 mandated test types** enumerated in coding-standards.md §8 (target: 100% test-type coverage per feature × platform cell, not a line-percentage).

2. Item lifecycle



Explanation (for readers/models that cannot see the diagram). A workable item begins `TODO`. The automatic multi-track ruler `[§11.4.187]` claims it exactly once through the append-only claim registry `[§11.4.176]`, moving it to `CLAIMED`; no second track may claim the same item or its hidden-coupling logical group. A headless worker then takes it to `IN_PROGRESS` on its own isolated git-worktree so concurrent tracks never collide in one checkout.

The middle of the lifecycle handles the two ways real work stalls or is judged. If an item hits an external dependency (hardware, a not-yet-merged prerequisite item, a long build) it goes `BLOCKED` and the track is auto-backfilled with other actionable work rather than idling `[§11.4.192/94]`; when the dependency clears it returns to `IN_PROGRESS`. When the reproduce-first RED test turns GREEN and every mandated test type passes, the item enters `REVIEW`, where an independent AI review runs on **Fable @ xhigh (Opus xhigh fallback)** `[§11.4.209]` — a genuine second opinion in a different model/context than the author.

The terminal transitions are the two review verdicts. A `NO-GO` sends the item back to `IN_PROGRESS` to iterate `[§11.4.134]` and be re-reviewed; a `GO` merges it onto the latest main with no force-push `[§11.4.113]` and the item is `DONE`. Because `REVIEW → DONE` is gated on `GO` and `IN_PROGRESS → REVIEW` is gated on all test types GREEN, an item can never reach `DONE` on a red test or an unreviewed diff — the state machine itself is the anti-bluff enforcement.

> Rendered PNG/SVG exported via Docs Chain (§11.4.65). Source: diagrams/item-lifecycle.mmd.

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3. Phase 1 — Foundation

3.1 Infrastructure (sub-phase 1.1)

ID	Title	Kind	Pri	Depends on	Gap
ATM-001	Repo + 4-upstream bootstrap; install `upstreams/*.sh`, git-hooks (`pre-commit` / `pre-push` / `post-commit` / `commit-msg`)	PRODUCTION-WIRE	P0	—	—
ATM-002	Rootless Podman Compose baseline via `vasic-digital/containers` (`pkg/boot` / `pkg/compose` / `pkg/health`)	PRODUCTION-WIRE	P0	ATM-001	—
ATM-003	`digital.vasic.database` wiring — SQLite dev / Postgres prod; `pkg/migration.Runner`	PRODUCTION-WIRE	P0	ATM-002	—
ATM-004	pgvector provisioning + `digital.vasic.vectordb` (pgvector backend, cosine) co-located in Postgres	PRODUCTION-WIRE	P0	ATM-003	3.1
ATM-005	`digital.vasic.observability` — OTel + Prometheus + logrus + ClickHouse + `pkg/health`	PRODUCTION-WIRE	P0	ATM-002	—
ATM-006	`vasic-digital/lets_encrypt` TLS per subdomain (HTTP-01/DNS-01, atomic deploy-hook + rollback)	PRODUCTION-WIRE	P1	ATM-002	—

ID	Title	Kind	Pri	Depends on	Gap
ATM-007	Dynamic ports — `digital.vasic.discovery` + `mdns` + `port_prefix` (deterministic ≤65535)	PRODUCTION-WIRE	P1	ATM-002	—
ATM-008	CodeGraph index of all own-org submodules `[§11.4.78-80]` + scheduled re-index	PRODUCTION-WIRE	P1	ATM-001	—
ATM-009	Docs Chain context (`.docs_chain/contexts/*.yaml`) — every `.md` gets HTML/PDF/DOCX siblings `[§11.4.65]`	PRODUCTION-WIRE	P1	ATM-001	10.1

ATM-004 acceptance (representative). pgvector extension present; a `VectorStore` cosine `<=>` search returns ids that hydrate against relational rows; the `HELIX_EMBEDDING_PROVIDER` is validated to be `llama` (never the `HashEmbedder` stub) in any RAG/search context — see `ATM-040` `[GAP: 2.1]`. Test types: unit, integration, performance (< 500 ms search SLO), benchmarking.

3.2 Core Services (sub-phase 1.2)

ID	Title	Kind	Pri	Depends on	Gap
ATM-010	**User Service** — three-tier RBAC (root/account-admin/user) on `auth` + `security/pkg/policy` + Catalogizer pattern	BUILD-NEW	P0	ATM-003, ATM-013	7.2, §11
ATM-011	**Event Bus service** — client-facing subscription wrapping `digital.vasic.eventbus` (NATS JetStream)	BUILD-NEW	P0	ATM-002	§11
ATM-012	**Asset Service** — decouple from `vasic-digital/Catalogizer` (multi-protocol, SQLCipher-at-rest, JWT+RBAC, WS, Range/`OpenSeekable`)	BUILD-NEW	P1	ATM-003, ATM-010	6.1, §11
ATM-013	`digital.vasic.auth` — add RS256/EdDSA signing + JWKS rotation (default is HMAC-SHA256)	EXTEND	P1	ATM-003	7.2
ATM-014	`digital.vasic.security` — sealed key store + **searchable-yet-sealed** credential representation (embed over redacted/tokenized form)	EXTEND	P2	ATM-003	7.1
ATM-015	`Security-KMP` — implement **native** secure storage (Android Keystore /	HARDEN	P0 (mobile)	—	7.3

ID	Title	Kind	Pri	Depends on	Gap
	iOS Keychain / Wasm) — currently in-memory **stub**				

ATM-010 acceptance. Root Admin bootstrapped owner-only at deploy; Root creates Accounts + Account-Admins; Account-Admins invite users; a user may belong to multiple Accounts; every permission check flows through the `security/pkg/policy` enforcer; TOTP MFA mandatory for admin tiers. Test types: unit, integration, e2e, security (authz matrix + privilege escalation), challenges, helixqa. Design plan: build-new-subsystems.md §User Service.

3.3 Integration (sub-phase 1.3)

ID	Title	Kind	Pri	Depends on	Gap
ATM-016	Herald — promote the `qaherald` `gotd/td` MTProto user client to a first-class Telegram thread-reader (forum topics `channels.getForumTopics`, replies `messages.getReplies`)	EXTEND	P0	ATM-003	5.1
ATM-017	**ThreadReader** abstraction submodule — root + organic reply-chain assembly, exclude system replies, resolve `access_hash`; shared across messengers	BUILD-NEW	P1	ATM-016	5.1, §11
ATM-018	**Max adapter** — Bot API (Go SDK) for bot scope **+** Go port of the OneMe user-WebSocket protocol for full channel/thread history	BUILD-NEW	P0	ATM-017	5.1, §11
ATM-019	Relational schema + migrations (posts, threads, replies, hashtags, categories, accounts, users, roles, permissions, memberships, assets, asset_links, processing_state, events, subscriptions, billing/metering, audit)	EXTEND	P0	ATM-003	3.2
ATM-020		EXTEND	P0	ATM-010, ATM-019	—

ID	Title	Kind	Pri	Depends on	Gap
	REST API skeleton — `/v1`, HTTP/3 (`vasic-digital/http3`) + H2 fallback, `auth`, `ratelimiter`, `security/pkg/headers`, `middleware`				
ATM-021	Real-time surface — `digital.vasic.streaming` WebSocket hub + SSE, bridged to the Event Bus service	EXTEND	P1	ATM-011, ATM-020	9

ATM-018 note `[OPEN: max-oneme-go-port]`. The OneMe reference implementations (`vkmax`, `max-mcp`, `MaxAPI`) are Python; the Go port is **unproven** and gated by a research spike (see build-new-subsystems.md §Max adapter). Bot-API scope ships first; full user-history scope is P0 but spike-gated.

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4. Phase 2 — Processing Engine

4.1 Content Acquisition (sub-phase 2.1)

ID	Title	Kind	Pri	Depends on	Gap
ATM-022	Acquisition poller (configurable cadence) + push triggers; assemble the **complete post** = root + organic reply chain	EXTEND	P0	ATM-016, ATM-019	5.1
ATM-023	Idempotent **single-claim per post** on `digital.vasic.background` (Postgres row/advisory lock) — exactly-once even under event storms	EXTEND	P0	ATM-011, ATM-019	2.9
ATM-024	Indirect hashtag determination (torrent→`Torrent` + `ToDownload`; git host→research; YouTube/media→download+research) + AI fallback	BUILD-NEW	P1	ATM-025	—

4.2 Processing Pipeline (sub-phase 2.2)

ID	Title	Kind	Pri	Depends on	Gap
ATM-025	**Skill-dispatch engine** — hashtag/content-type → Skill(s), ordered download→convert→analyze→research→reply, over BackgroundTasks	BUILD-NEW	P0	ATM-011, ATM-023, ATM-036	4.1
ATM-026	Multi-hashtag additive resolution + precedence table (download > convert > analyze > research > reply) + `SortOrder`	EXTEND	P0	ATM-025	2.9
ATM-027	Retry/back-off (exp + jitter, max 5, base 2 s, cap 5 min) + circuit-breaker; per-post soft timeout	EXTEND	P0	ATM-025	—
ATM-028	**Download Manager** submodule — HTTP/1.1/2/3+QUIC+Brotli + reuse `filesystem` for FTP/SMB/NFS/WebDav; queue, resumable+segmented, progress, retry, callback	BUILD-NEW	P0	ATM-029, ATM-030	6.2, 6.3, §11
ATM-029	`digital.vasic.filesystem` — add HTTP(S) source protocol; fix NFS non-Linux platform listing; expose `OpenSeekable` uniformly for Range	EXTEND	P1	—	6.2
ATM-030	**Standardized callback/task module** — accept-task→async→status→callback→error→retry; common schema (job id, state, progress, result-asset-ref, error)	BUILD-NEW	P1	ATM-011	6.6, §11
ATM-031	MeTube — add **outbound completion webhook** (poll-only today) + Asset Service integration + `…-web` conversion profiles	EXTEND	P0	ATM-012, ATM-030	6.5
ATM-032	Boba — standardize the existing SSE + `POST /api/v1/hooks` callback to the shared schema; Asset Service integration	EXTEND	P1	ATM-030	6.4
ATM-033	**OCR adapter** — Tesseract (cgo `gosseract` / subprocess) + PaddleOCR option behind a `VisionEngine` `OCRProvider` seam; per-word boxes; hybrid pipeline	BUILD-NEW	P0	—	2.6, §11
ATM-034	Media transcode profiles — raw preserved + `…-web` H.264/AAC fMP4 (+H.265/AV1), 1080/720/480 HLS+DASH; audio MP3 320k/Opus 128k/FLAC	EXTEND	P1	ATM-012, ATM-031	6.1
ATM-035	Post-processing — status reply to original post (Robot/User acct), fire events,	EXTEND	P0	ATM-025, ATM-039	—

ID	Title	Kind	Pri	Depends on	Gap
	embed+index, grow Skill-Trees; **never process own replies**				

ATM-033 note `[GAP: 2.6]`. VisionEngine has **no OCR engine** (grep-confirmed). This item adds one behind a first-class `OCRProvider` seam and wires `TextRegion` to a real recognizer. Design plan: build-new-subsystems.md §OCR adapter.

4.3 Skills Integration (sub-phase 2.3)

ID	Title	Kind	Pri	Depends on	Gap
ATM-036	Author a Skill (recipe) for **every** documented content type — all categories in parallel `[OPERATOR]`	EXTEND	P0	—	4.1
ATM-037	Standardize the Skill file format (one canonical `SKILL.md` YAML-frontmatter schema; migrate no-frontmatter dirs); fix the `catalog-engine` “Test Catalog” mislabel; deterministic `INDEX.md` regen	HARDEN	P1	ATM-036	4.1
ATM-038	Triage + burn down the Skill-Graph MVP **95 open** findings relevant to Thready	AUDIT	P1	ATM-037	4.1

4.4 Semantic Search (sub-phase 2.4)

ID	Title	Kind	Pri	Depends on	Gap
ATM-039	**Semantic-search service** (Lumen-style, in-house) — `embeddings` + `vectordb` + `MCP_Module` over llama.cpp/HelixLLM `/v1/embeddings`; index posts **and** generated materials	BUILD-NEW	P0	ATM-004, ATM-040	2.7, §11
ATM-040	HelixLLM — enforce `HELIX_EMBEDDING_PROVIDER=llama`; **fail loudly** (not warn) if the non-semantic `HashEmbedder` is selected in RAG/search; load a real code-tuned embedding GGUF	HARDEN	P0	—	2.1
ATM-041	`digital.vasic.embeddings` — first-class llama.cpp/HelixLLM provider with health checks + dimension discovery; parameterize embedding dimension (remove the 768 hardcode)	EXTEND	P1	ATM-040	2.1, 2.7
ATM-042	`digital.vasic.vectordb` — harden + integration-test the **Qdrant** backend to full pgvector parity; ANN index tuning + benchmark for < 500 ms SLO	HARDEN	P1	ATM-004	3.1
ATM-043	`digital.vasic.database` — time-partitioning + retention/archive	EXTEND	P1	ATM-003	3.2

ID	Title	Kind	Pri	Depends on	Gap
	helpers for 10k+/ day posts; validate `storage` MinIO signed-URL parity; pool + pgvector co-location tuning				

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5. Phase 3 — Client Applications

ID	Title	Kind	Pri	Depends on	Gap
ATM-044	Angular 19 product portal on the shared OpenDesign `design_system` (Web + CLI first `[OPERATOR]`)	EXTEND	P0	ATM-020, ATM-060	8.1
ATM-045	Tauri 2 desktop (Rust core + Angular UI) — org standard	EXTEND	P1	ATM-044	—
ATM-046	Native mobile clients (Compose/ Android, SwiftUI/ iOS, ArkTS/ HarmonyOS, Qt/ Aurora + `helix_shims`) + KMP shared logic	BUILD-NEW	P2	ATM-047, ATM-015	8.3, 8.4, 8.5
ATM-047	KMP module fleet — add CI + Maven publish + shared convention plugin; implement `Database-KMP` with SQLDelight (interfaces-only today)	HARDEN	P1	—	8.4
ATM-048	Go/Cobra **CLI** (headless) sharing the SDK with the TUI — first alongside Web `[OPERATOR]`	EXTEND	P0	ATM-060	—
ATM-049	Bubble Tea + Cobra + Lipgloss **TUI** (as `helix_track_cli`)	EXTEND	P1	ATM-048	8.6
ATM-050	Thready brand theme from `Logo.png` on the helix-green base; mature `design_system`; publish `@vasic-`	EXTEND	P1	—	8.1

ID	Title	Kind	Pri	Depends on	Gap
	`digital/design-system`				
ATM-051	`helix_design` — implement per-platform token packages (Flutter, Qt/Aurora, CSS) from the OpenDesign source (currently a scaffold)	HARDEN	P1	ATM-050	8.2

ATM-046 note. Only path to HarmonyOS + Aurora is native clients + `helix_shims`, currently **scaffolds** `[GAP: 8.5]`; `Security-KMP` mobile storage must be real (`ATM-015`) before any mobile release. Mobile is P2 per the Web+CLI-first operator decision.

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6. Phase 4 — Testing & Deployment

ID	Title	Kind	Pri	Depends on	Gap
ATM-052	Unit + `go-mutesting` mutation testing + **paired-mutation anti-bluff gates** (real behavior, not stubs)	EXTEND	P0	per-item	12
ATM-053	Integration — all cross-module contracts + critical paths first; no-fakes-beyond-unit	EXTEND	P0	Phase 1–2	§11.4.27
ATM-054	System/e2e + **HelixQA** YAML banks (runtime evidence) + **Challenges** scenario banks	EXTEND	P0	ATM-053	9.1, 9.3
ATM-055	Security — SonarQube (CLI + rootless-Podman server) + Snyk + fuzzing + CVE + DDoS/authz	EXTEND	P0	ATM-053	7.1
ATM-056	Performance/ benchmarking/ stress/scaling/ chaos — validate SLOs (API p95 < 150 ms, search < 500 ms) + RPO/RTO	EXTEND	P1	ATM-053	3.1, 3.2
ATM-057	Production deployment — 3 isolated env stacks (`dev.` / `sta.` / `thready.`), rootless Podman Compose, Let's Encrypt, backup/DR runbook (RPO ≈ 1 h, RTO ≈ 4 h)	EXTEND	P0	ATM-002, ATM-006	—

7. Cross-cutting items

These span all phases (see the dotted band in index.md §4).

ID	Title	Kind	Pri	Gap
ATM-058	**Anti-bluff sweep** — every `SCAFFOLD` / `DESIGN-ONLY` dep gets a paired-mutation gate before reliance	AUDIT	P1	12
ATM-059	**Decoupling audit** `[§11.4.28]` — each reused submodule verified project-not-aware/config-injected; no nested own-org chains	AUDIT	P1	12
ATM-060	SDK codegen — OpenAPI 3.1 + Protobuf via the `helix_proto` pattern (`buf` → Go/Rust; `openapi-generator` → TS/Dart) + thin idiomatic layer per language	BUILD-NEW	P1	13
ATM-061	`TOON` + `token_optimizer` — implement (load-bearing for the token-optimization mandate `[§11.4.198]`) **or** explicitly mark out-of-scope for MVP	HARDEN	P1	2.9
ATM-062	`session_orchestrator` — **rescoped (Pass 3):** the atomic track-claim registry is **now shipped at source** (`vasic-digital/session_orchestrator/claim/claim.go` — exactly-once `Registry.TryClaim` / `Release`), superseding the gap register's DESIGN-ONLY status; item is **wire+verify+integrate**, not build-from-scratch `[VERIFIED-SOURCE]`	HARDEN	P1	2.9
ATM-063	HelixAgent — resolve the HelixAgent↔HelixCode identity/module split; enumerate + close admitted stub/bluff areas; pin only the packages Thready needs	HARDEN	P2	2.2

ID	Title	Kind	Pri	Gap
ATM-064	LLMProvider — per-adapter contract tests for every adapter Thready relies on; confirm embeddings interface location	AUDIT	P1	2.3, 13
ATM-065	LLMsVerifier — reconcile the `:7061` vs `:8080` port; expose a stable scoring API for HelixLLM's fallback chain; prune doc sprawl	HARDEN	P2	2.5
ATM-070	`digital.vasic.memory` — **decision (recorded, not deferred silently):** Thready standardizes on the `vectordb` + `embeddings` + `rag` stack (via the Semantic-search service `ATM-039`) and does **not** rely on `memory`'s word-overlap (Jaccard) search; if `memory` is later adopted, back it with `vectordb` for real semantic recall. `HelixMemory` (4 external services: Mem0/Cognee/Letta/Graphiti) is **deferred** unless full cognitive memory is required	AUDIT	P2	2.8
ATM-071	`HelixDevelopment/HelixStream` — **defer** (SCAFFOLD, ~5 commits): not on the Thready MVP critical path; if streaming-app testing is later required, harden the scaffold and add CI before reliance	AUDIT	P2	9.2

8. Open items register

Tracked `[OPEN: …]` items, each carried as a workable item so nothing is papered over.

ID	`[OPEN: …]`	Why unresolved	Resolution plan
ATM-066	`constitution-anchor-verify`	Local Constitution submodule copy tops out at §11.4.192; **§11.4.196/198/209** wording taken from the final request's descriptions, not read at source	Re-verify against the canonical `HelixDevelopment/helix_constitution` before relying on precise normative text; update citations
ATM-018/ ATM-067	`max-oneme-go-port`	OneMe user-WS reference impls are Python; Go port unproven	Research spike + protocol capture; ship Bot-API scope first (see build-new plan)
ATM-067	~~`agentpool-contract`~~ **CLOSED (Pass 3)**	Was not locally cloned; now **read at source** — `vasic-digital/LLMOrchestrator/pkg/agent/pool.go`: `AgentPool.Acquire(ctx, AgentRequirements) (Agent, error)` + `Release(Agent)`, capability-matched via `Agent.Capabilities()` `AgentCapabilities` `[VERIFIED-SOURCE]`	**Done** — contract documented in workable-items-detail.md §ATM-067 + agent-orchestration.md §3. Residual (non-blocking): add HarmonyOS/ Aurora build-agent capabilities if those become agent tasks
ATM-068	`flagged-modules-verify` **(partial, Pass 3)**	`vasic-digital/Agentic` now **verified to exist** at source ("Graph-based agentic workflow orchestration"); `Normalize` / `conversation` / `SkillRegistry` / `ToolSchema` / `MCP_Module` / `Planning` / `AgentWrapper` still **not** individually source-verified	Source-verify + pin exact import paths for the residual seven before any item depends on them `[GAP: 2.9, §13]`
ATM-069	`docs-chain-tooling`	Docs Chain honest-SKIPs md→HTML/ PDF when `pandoc` / `weasyprint` are absent (as on the current host)	Provision pandoc/weasyprint on dev/CI hosts so `ATM-009` siblings generate `[GAP: 10.1]`
ATM-072	`sibling-area-index-missing`	The `../architecture/index.md` and `../database/index.md` canonical entry points `[§11.4.212]` do not yet exist, so the upstream cross-links from index.md §3 resolve only once those areas add their `index.md` (every other sibling area — api/testing/ deployment/design/user-guides — already has one)	Architecture and Database area owners create their `index.md`; until then the links target the canonical path, tracked here so the dead reference is not papered over `[CONVENTIONS §5/§7]`

9. Gap-register coverage matrix

Every gap-register subsystem relevant to Development & Orchestration is covered by ≥1 item. (Data subsystems whose gaps are addressed by build/harden items are included; purely runtime/architecture gaps are cross-referenced to their owning area.)

Gap-register §	Headline	Covered by
2.1 HelixLLM	HashEmbedder stub / 768 hardcode / verifier port	ATM-040, ATM-041, ATM-065
2.2 HelixAgent	Identity blur + residual stubs	ATM-063
2.3 LLMProvider	40+ adapters unaudited	ATM-064
2.4 LLMOrchestrator	`AgentPool` capability-match contract not source-verified	ATM-067
2.5 LLMsVerifier	Port discrepancy	ATM-065
2.6 VisionEngine	No OCR engine	ATM-033
2.7 Embeddings	No native llama.cpp backend	ATM-041, ATM-039
2.8 Memory / HelixMemory	Word-overlap search; HelixMemory heavyweight	ATM-070 (standardize on vectordb+embeddings+rag; defer HelixMemory)
2.9 agent-support	TOON/token_optimizer design-only; `session_orchestrator` **now shipped** (claim registry read at source — ATM-062 rescoped to wire/verify); FLAGGED modules (`Agentic` verified, seven residual)	ATM-061, ATM-062, ATM-068
3.1 VectorDB	Qdrant/others unverified	ATM-042
3.2 database/storage	No partitioning; MinIO signed-URL parity	ATM-043
4.1 helix_skills	No execution engine; format inconsistency; 95 findings	ATM-025, ATM-037, ATM-038, ATM-036
5.1 herald	MTProto trapped in QA harness; Max empty stub; no ThreadReader	ATM-016, ATM-017, ATM-018
6.1 Catalogizer→Asset	Not decoupled	ATM-012, ATM-034
6.2 filesystem	No HTTP source; no DL semantics; NFS listing	ATM-029
6.3 Download Manager	Does not exist	ATM-028
6.4 Boba	Bespoke callback	ATM-032
6.5 MeTube	No outbound webhook	ATM-031
6.6 callback module	Not extracted	ATM-030

Gap-register §	Headline	Covered by
7.1 security	Wire scanner adapters; searchable-sealed creds	ATM-014, ATM-055
7.2 auth	HMAC default; no RBAC	ATM-013, ATM-010
7.3 Security-KMP	Mobile storage in-memory stub	ATM-015
7.4 Auth-KMP	Must consume the fixed Security-KMP for token storage	ATM-046 (consumes hardened `Security-KMP`, `ATM-015`)
8.1 design_system	Standalone maturation	ATM-050
8.2 helix_design	Non-web arm scaffold	ATM-051
8.3 helix_ui	Flutter UI scaffold — conditional	ATM-046 (only if the Flutter family is chosen; the native ArkTS/Qt path is the default for HarmonyOS/Aurora reach)
8.4 KMP fleet	No CI/publish; Database-KMP interfaces-only	ATM-047
8.5 helix_shims/native	HarmonyOS/Aurora scaffolds	ATM-046
8.6 TS libs/track_cli	No deep audit	ATM-049
9.1 HelixQA	YAML banks + mandatory runtime evidence	ATM-054
9.2 HelixStream	SCAFFOLD — deferred (decision recorded)	ATM-071
9.3 Panoptic/VisualRegression	scenario banks + CI for the VR family	ATM-054, ATM-056
9.4 DocProcessor	feature-map → docs↔tests coverage	ATM-054
9.x test execution	the 15 test-type banks run locally / via the fleet (no server CI)	ATM-052…ATM-056
10.1 docs_chain	Pandoc/weasyprint SKIP	ATM-009, ATM-069
§11 New subsystems	Asset/Download/Max/OCR/User/callback/EventBus/ThreadReader/Semantic-search	ATM-010, ATM-011, ATM-012, ATM-017, ATM-018, ATM-028, ATM-030, ATM-033, ATM-039
12 cross-cutting	anti-bluff / decoupling / CI-equiv	ATM-058, ATM-059
13 Re-verification backlog		

Gap-register §	Headline	Covered by
	FLAGGED adapters/modules + constitution anchors source-verified before reliance	ATM-064, ATM-066, ATM-067, ATM-068

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